# Lecture 5: Single Point Positioning

GNSS should provide:

- position
- velocity
- time

Navigation solution: Single Point Positioning (SPP)

Measurement Model for code observations

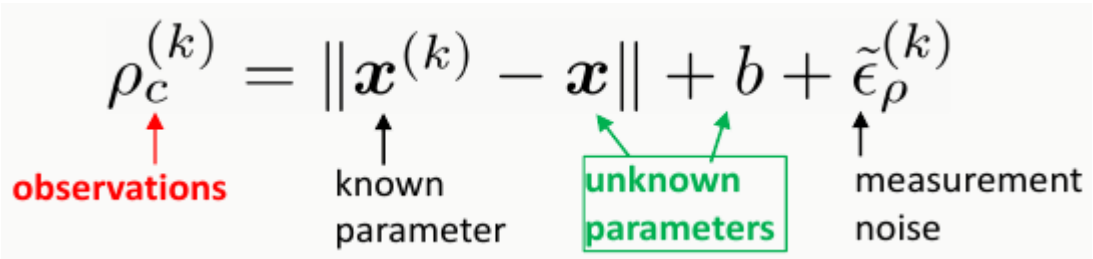
$$\rho = r + I_\rho + T_\rho + c[\delta t_u - \delta t^s] + \varepsilon_\rho$$

- Apply **corrections** to the observation
  - From navigation message:
    - Satellite clock offset
    - Relativistic effect
    - Ionospheric delay
  - Troposheric delay -> small with respect to other errors
- Simplified measurement model:
- $\rho_c^{(k)} = r^{(k)} + c\delta t_u + \tilde{\varepsilon}_\rho^{(k)}$ with k: satellites
- Geometric distance between user and satellites
  - $r^{(k)} = ||\boldsymbol{x}^{(k)} - \boldsymbol{x}||$

  Measurement model for $k \geq 4$:

$$\rho_c^{(k)} = ||\boldsymbol{x}^{(k)} - \boldsymbol{x}|| + c\delta t_u + \tilde{\varepsilon}_\rho^{(k)}$$

Functional Relationship between **observations** and **parameters** to estimate (for every satellite k)




- **Goal**: Estimate unknown parameters based on a set of observations
- **Method**: parameter adjustment(least squares)

- General linear case

$$\boldsymbol{\rho} = \mathbf{G}\mathbf{x}$$

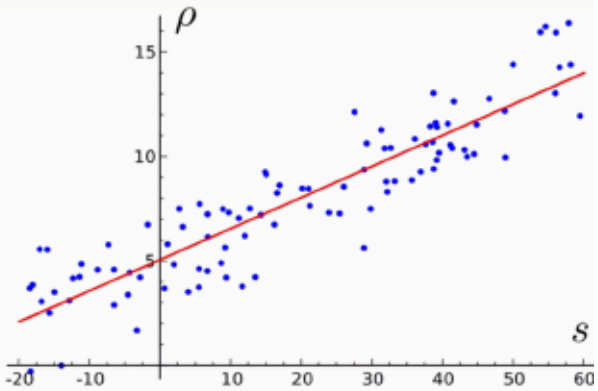
vector of observations

vector of unknown parameters

matrix describing the relationship between unknown parameters and observations (Design Matrix)

Example Line Fit

$$\begin{bmatrix} \rho_1 \\ \rho_2 \\ \vdots \\ \rho_n \end{bmatrix} = \begin{bmatrix} s_1 & 1 \\ s_2 & 1 \\ \vdots & \\ s_n & 1 \end{bmatrix} \begin{bmatrix} x \\ b \end{bmatrix}$$




- Solution (assuming all observations have the same uncertainty)

$$\boldsymbol{\rho} = \mathbf{G}\mathbf{x} \qquad \mathbf{P} = \boldsymbol{\Sigma}_{ll}^{-1}$$

covariance matrix of observations

$$\begin{bmatrix} \sigma_{\rho,1}^2 & 0 & \dots & 0 \\ 0 & \sigma_{\rho,2}^2 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & \sigma_{\rho,n}^2 \end{bmatrix}$$

1) Define error function $\mathbf{e} = \mathbf{G}\mathbf{x} - \boldsymbol{\rho}$
2) Find minimum of $\mathbf{e}^T\mathbf{P}\mathbf{e}$ by building the derivative and setting it to zero

$$\hat{\mathbf{x}} = \left(\mathbf{G}^T\mathbf{P}\mathbf{G}\right)^{-1}\mathbf{G}^T\mathbf{P}\boldsymbol{\rho}$$

**Problem**: Functional relationship is non-linear

**Solution**: Linearisation

- Linearising about an approximate solution $\boldsymbol{x}_0 = (x_0, y_0, z_0), b_0$
- Corresponding approximate observation $\rho_0^{(k)} = ||\boldsymbol{x}^{(k)} - \boldsymbol{x}_0|| + b_0$
- True position and bias
  - $\boldsymbol{x} = \boldsymbol{x}_0 + \delta\boldsymbol{x}$ & $\quad b = b_0 + \delta b$

- True observation $\rho_c^{(k)} = \rho_0^{(k)} + \delta\rho^{(k)}$

**Parameter Adjustment**

$$\rho_c^{(k)} = \rho_0^{(k)} + \delta\rho^{(k)} \qquad \mathbf{x} = \mathbf{x}_0 + \delta\mathbf{x} \qquad b = b_0 + \delta b$$

- Rewrite $\delta\rho^{(k)}$ as (linear) function of $\delta\mathbf{x}, \delta b$

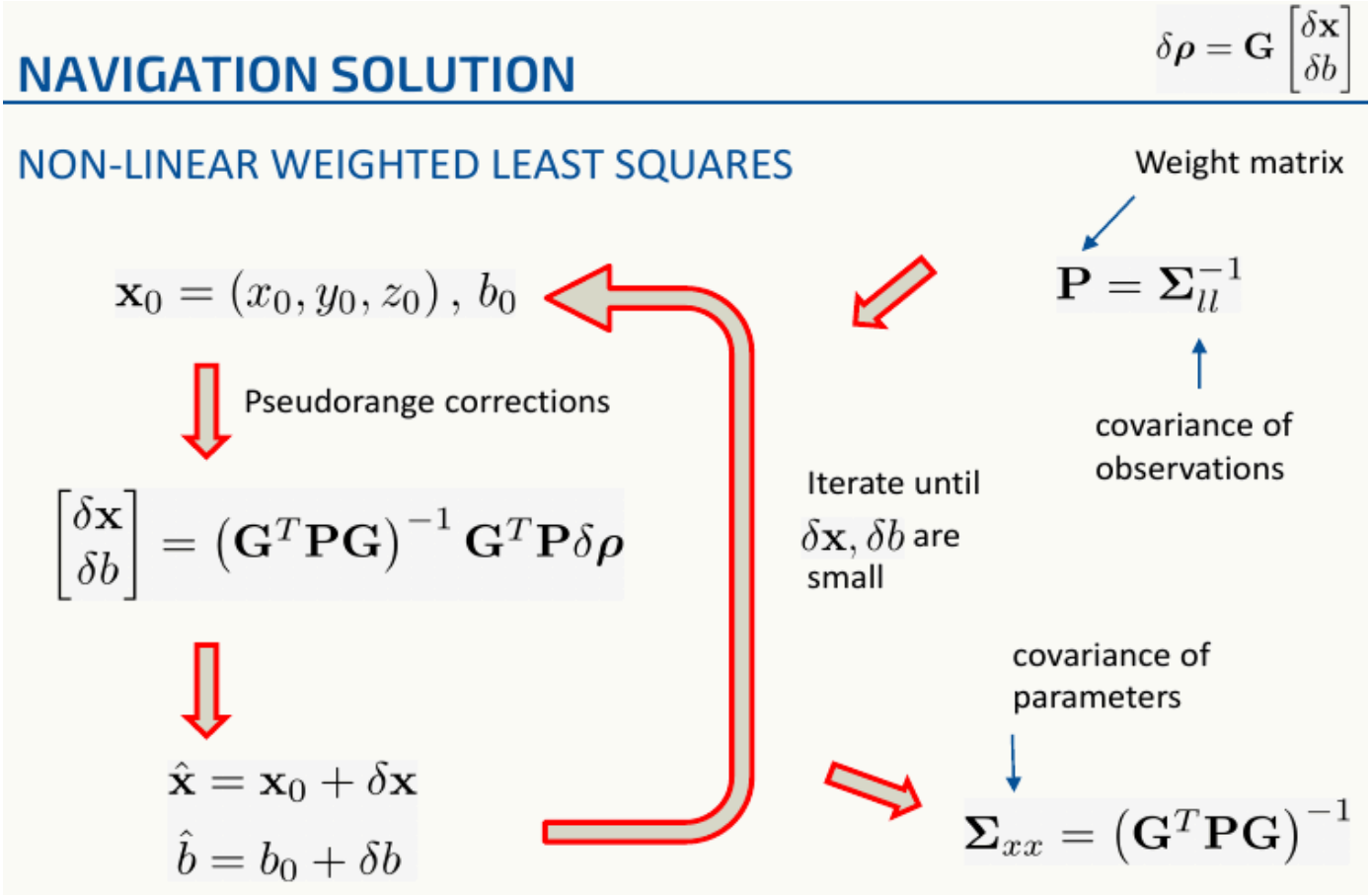
$$\begin{aligned}
\delta\rho^{(k)} &= \rho_c^{(k)} - \rho_0^{(k)} \\
&= ||\mathbf{x}^{(k)} - \mathbf{x}_0 - \delta\mathbf{x}|| - ||\mathbf{x}^{(k)} - \mathbf{x}_0|| + (b - b_0) + \tilde{\epsilon}_\rho^{(k)} \\
&\approx \boxed{\frac{(\mathbf{x}^{(k)} - \mathbf{x}_0)}{||\mathbf{x}^{(k)} - \mathbf{x}_0||}} \cdot \delta\mathbf{x} + \delta b + \tilde{\epsilon}_\rho^{(k)} \\
&= \boxed{-\mathbf{1}^{(k)}} \cdot \delta\mathbf{x} + \delta b + \tilde{\epsilon}_\rho^{(k)}
\end{aligned}$$

Taylor series expansion of norm

Line of sight vector from approximated user position to the satellite

$$\delta\boldsymbol{\rho} = \begin{bmatrix} \delta\rho_c^{(1)} \\ \delta\rho_c^{(2)} \\ \dots \\ \delta\rho_c^{(K)} \end{bmatrix} = \begin{bmatrix} (-\mathbf{1}^{(1)})^T & 1 \\ (-\mathbf{1}^{(2)})^T & 1 \\ \dots & \\ (-\mathbf{1}^{(K)})^T & 1 \end{bmatrix} = \begin{bmatrix} \delta\boldsymbol{x} \\ \delta b \end{bmatrix} + \boldsymbol{\varepsilon}_\rho = \boldsymbol{G} \begin{bmatrix} \delta\boldsymbol{x} \\ \delta b \end{bmatrix} + \boldsymbol{\varepsilon}_\rho$$

- K linear equations for four unknown $\delta\boldsymbol{x}, \delta b$ and observations $\delta\boldsymbol{\rho}$
- **Geometry matrix G contains only approximate positions and satellite coordinates**
- Can be solved for 4 observations, if $rank(G) = 4$
- Can be solved for over-determined case $k > 4$ using least-squares approach




### Stochastic Model

- How does P look like?
- Stochastic model for GNSS is not well known
  - Variances are different for different pseudoranges (e.g. elevation dependant)
  - pseudoranges may be correlated
    - Iono & tropo models act on all observations in the same way
- Consideration of correlations is very difficult
- Use of a simplified stochastic model without correlations

## STOCHASTIC MODEL

- Summary of errors

| *Error Source* | *RMS Range Error (RE)* |
|---|---|
| Satellite clock and ephemeris parameters | $\sigma_{RE/CS} \approx 3$ m |
| Atmospheric propagation modeling | $\sigma_{RE/P} \approx 5$ m |
| Receiver noise and multipath | $\sigma_{RE/RNM} \approx 1$ m |
| User Range Error (URE) | $\sigma_{URE} \approx 6$ m |

- User Range Error (URE) $\sigma_{URE} = \sqrt{\sigma^2_{RE/CS} + \sigma^2_{RE/P} + \sigma^2_{RE/RNM}}$
- **Simplified stochastic model**

$$P = 1/\sigma^2_{URE} \cdot \boldsymbol{I}$$

unity matrix or
elevation dependant weighting $sin(E^{(k)})$

- Non linear observation model $\rho_c^{(k)} = \|\boldsymbol{x}^{(k)} - \boldsymbol{x}\| + b + \tilde{\epsilon}_\rho^{(k)}$
- Approximation $\mathbf{x} = \mathbf{x}_0 + \delta\mathbf{x}$ $b = b_0 + \delta b$ $\rho_c^{(k)} = \rho_0^{(k)} + \delta\rho^{(k)}$
- Linearization

$$\delta\boldsymbol{\rho} = \begin{bmatrix} \delta\rho_c^{(1)} \\ \delta\rho_c^{(2)} \\ \vdots \\ \delta\rho_c^{(K)} \end{bmatrix} = \begin{bmatrix} \left(-\mathbf{1}^{(1)}\right)^T & 1 \\ \left(-\mathbf{1}^{(2)}\right)^T & 1 \\ \vdots & \\ \left(-\mathbf{1}^{(K)}\right)^T & 1 \end{bmatrix} \begin{bmatrix} \delta\mathbf{x} \\ \delta b \end{bmatrix} + \boldsymbol{\epsilon}_\rho = \mathbf{G} \begin{bmatrix} \delta\mathbf{x} \\ \delta b \end{bmatrix} + \boldsymbol{\epsilon}_\rho$$

- Solve iteratively using weighed least squares

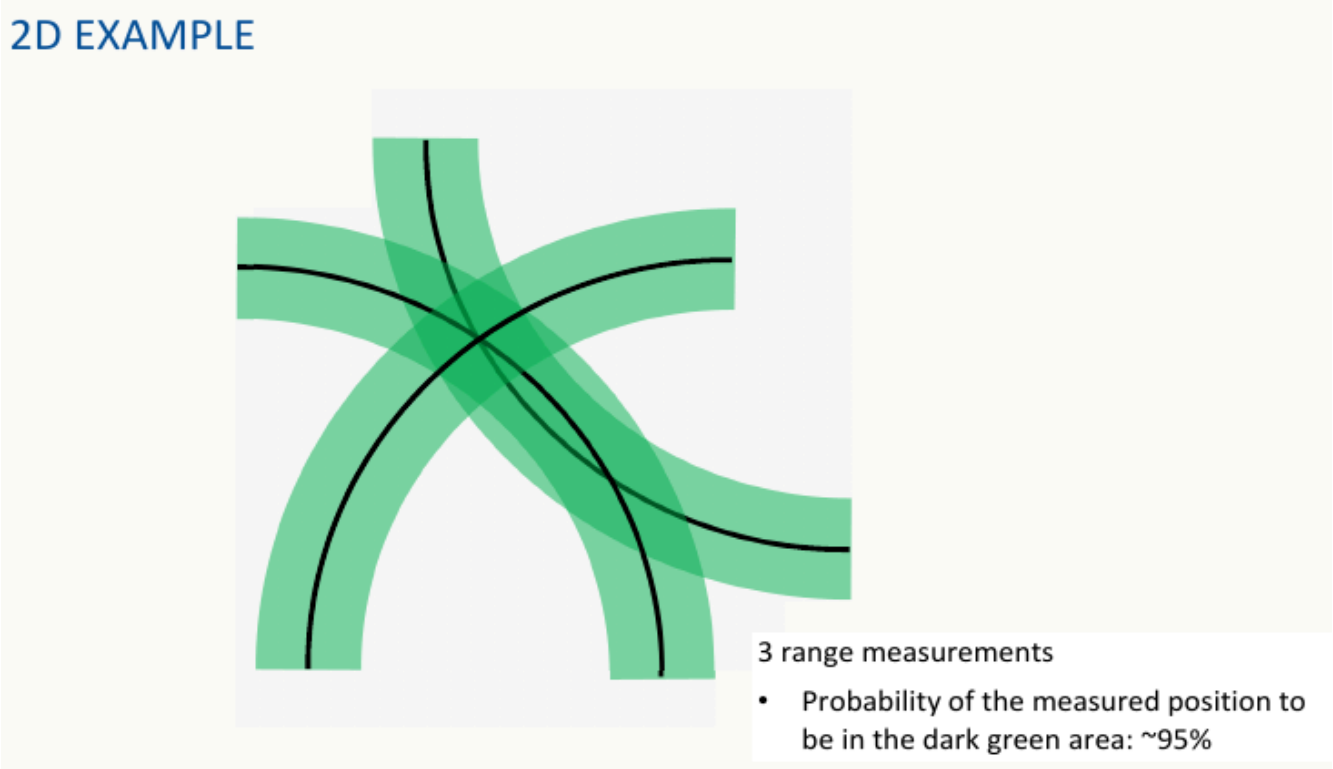
$$\begin{bmatrix} \delta\mathbf{x} \\ \delta b \end{bmatrix} = \left(\mathbf{G}^T\mathbf{P}\mathbf{G}\right)^{-1}\mathbf{G}^T\mathbf{P}\delta\boldsymbol{\rho}$$

$$\mathbf{P} = \boldsymbol{\Sigma}_{ll}^{-1} = \frac{1}{\sigma^2_{\mathrm{URE}}}\mathbb{1}_3$$ Weights

$$\boldsymbol{\Sigma}_{xx} = \left(\mathbf{G}^T\mathbf{P}\mathbf{G}\right)^{-1}$$ Covariance of parameters

## Satellite Geometry

- True range = black circle
- Stochastic + systematic errors -> measured range is somewhere in the green area
- width depends on the URE $\sigma$

- 


## Dilution of precision

- Covariance of the parameter estimation

$$\boldsymbol{\Sigma}_{xx} = cov \begin{bmatrix} \delta \boldsymbol{x} \\ \delta b \end{bmatrix} = (\boldsymbol{G}^T \boldsymbol{P} \boldsymbol{G})^{-1} = \sigma_{URE}^2 (\boldsymbol{G}^{\boldsymbol{T}} \boldsymbol{G})^{-1} = \sigma_{URE}^2 \boldsymbol{H}$$

with $\boldsymbol{P} = 1/\sigma_{URE}^2 \cdot \boldsymbol{I}$
and $\boldsymbol{H}$ contains only information about the user-satellite geometry!

$$\boldsymbol{H} = \begin{bmatrix} H_{xx} & H_{xy} & H_{xz} & H_{xb} \\ H_{yx} & H_{yy} & H_{yz} & H_{yb} \\ H_{zx} & H_{zy} & H_{zz} & H_{zb} \\ H_{bx} & H_{by} & H_{bz} & H_{bb} \end{bmatrix}$$

- Geometric Dilution of Precision $GDOP = \sqrt{H_{xx} + H_{yy} + H_{zz} + H_{bb}}$
- Position Dilution of Precision $PDOP = \sqrt{H_{xx} + H_{yy} + H_{zz}}$
- Time Dilution of Precision $TDOP = \sqrt{H_{bb}}$
- 3D position error $RMS = \sigma_{URE} \cdot PDOP$
- Clock bias error $RMS = \sigma_{URE} \cdot TDOP$
- Position and clock bias error $RMS = \sigma_{URE} \cdot GDOP$

- Transformation to local tangential system (East North Up)

$$cov\begin{bmatrix}\Delta E\\ \Delta N\\ \Delta U\end{bmatrix} = \sigma_{URE}^2 \mathbf{R}_L \mathbf{H}_{[x\ldots z,x\ldots z]}\mathbf{R}^T = \sigma_{URE}^2 \begin{bmatrix} EDOP^2 & & \\ & NDOP^2 & \\ & & VDOP^2 \end{bmatrix}$$

$$\mathbf{R}_L = \begin{bmatrix} -\sin\lambda & -\sin\varphi\cos\lambda & \cos\varphi\cos\lambda \\ \cos\lambda & -\sin\varphi\sin\lambda & \cos\varphi\sin\lambda \\ 0 & \cos\varphi & \sin\varphi \end{bmatrix}$$

$\lambda, \varphi$ : Longitude and Latitude (~ Position)

$HDOP = \sqrt{EDOP^2 + NDOP^2}$ ... horizontal dilution of precision

➔ horizontal error $RMS = \sigma_{URE} \cdot HDOP$

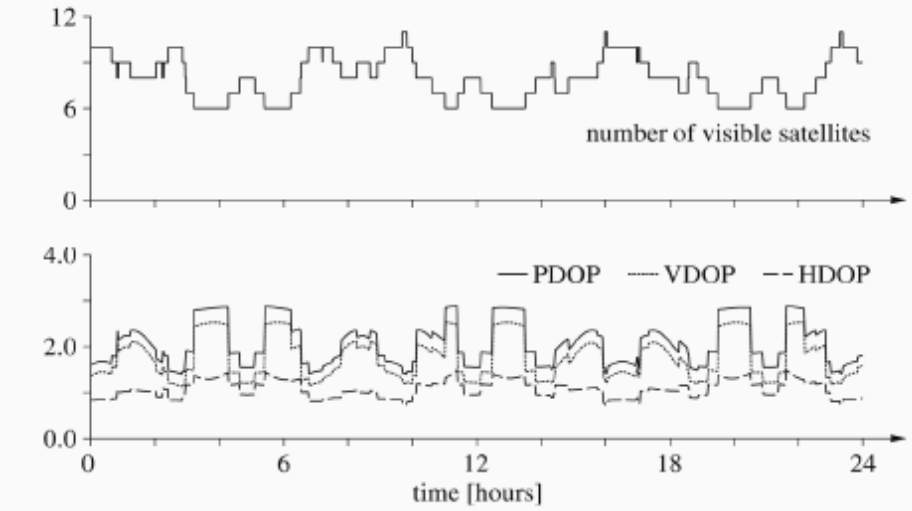
$VDOP$ ... vertical dilution of precision

➔ vertical error $RMS = \sigma_{URE} \cdot VDOP$

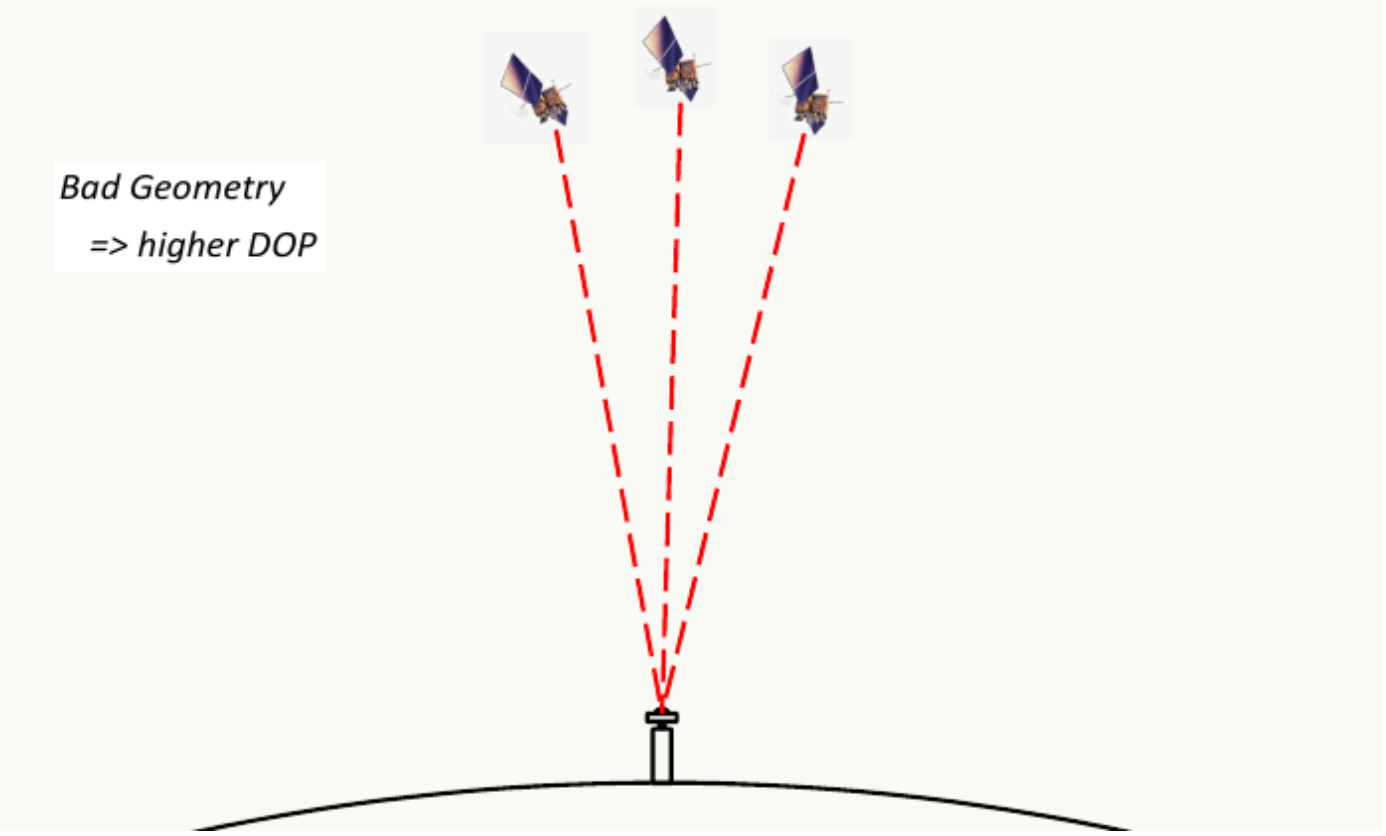
## DILUTION OF PRECISION

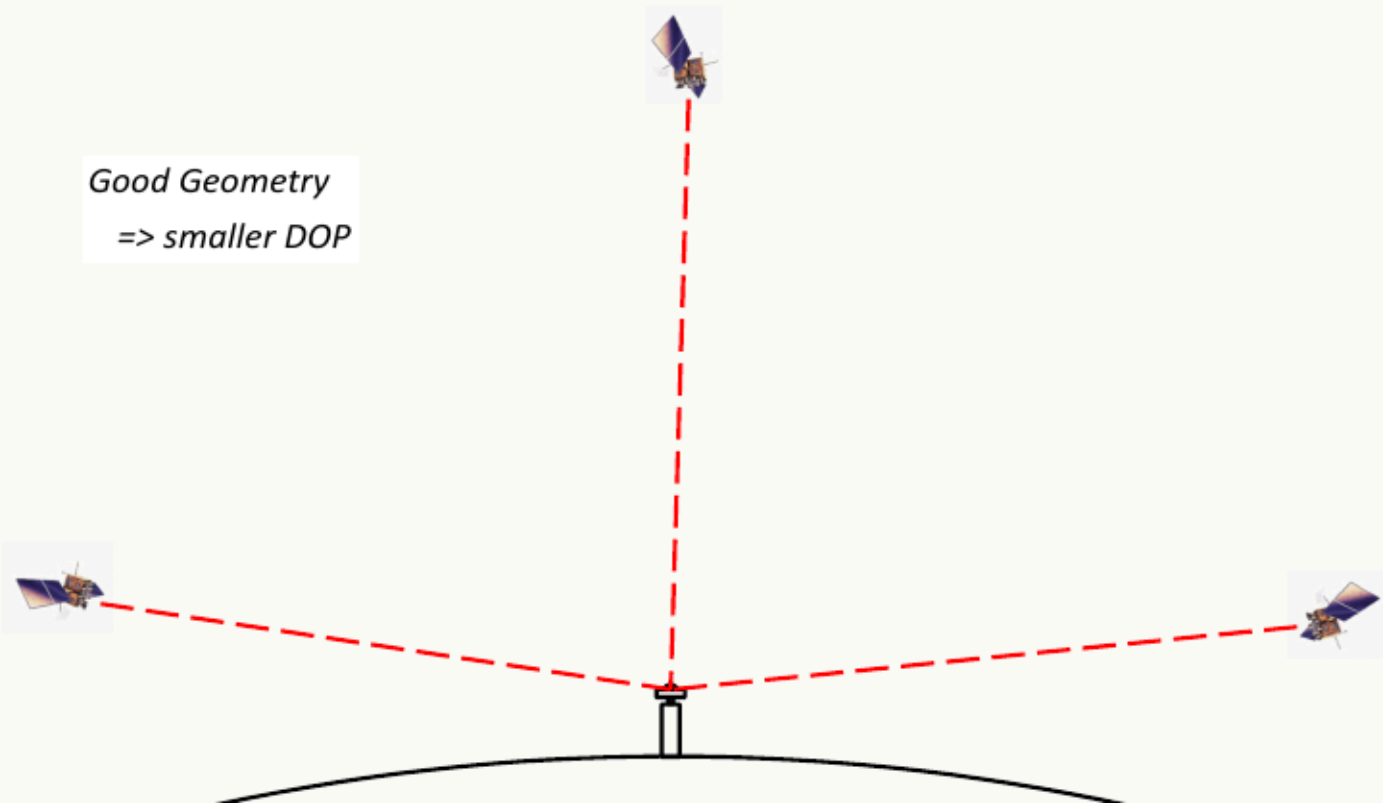
- Typical DOP values

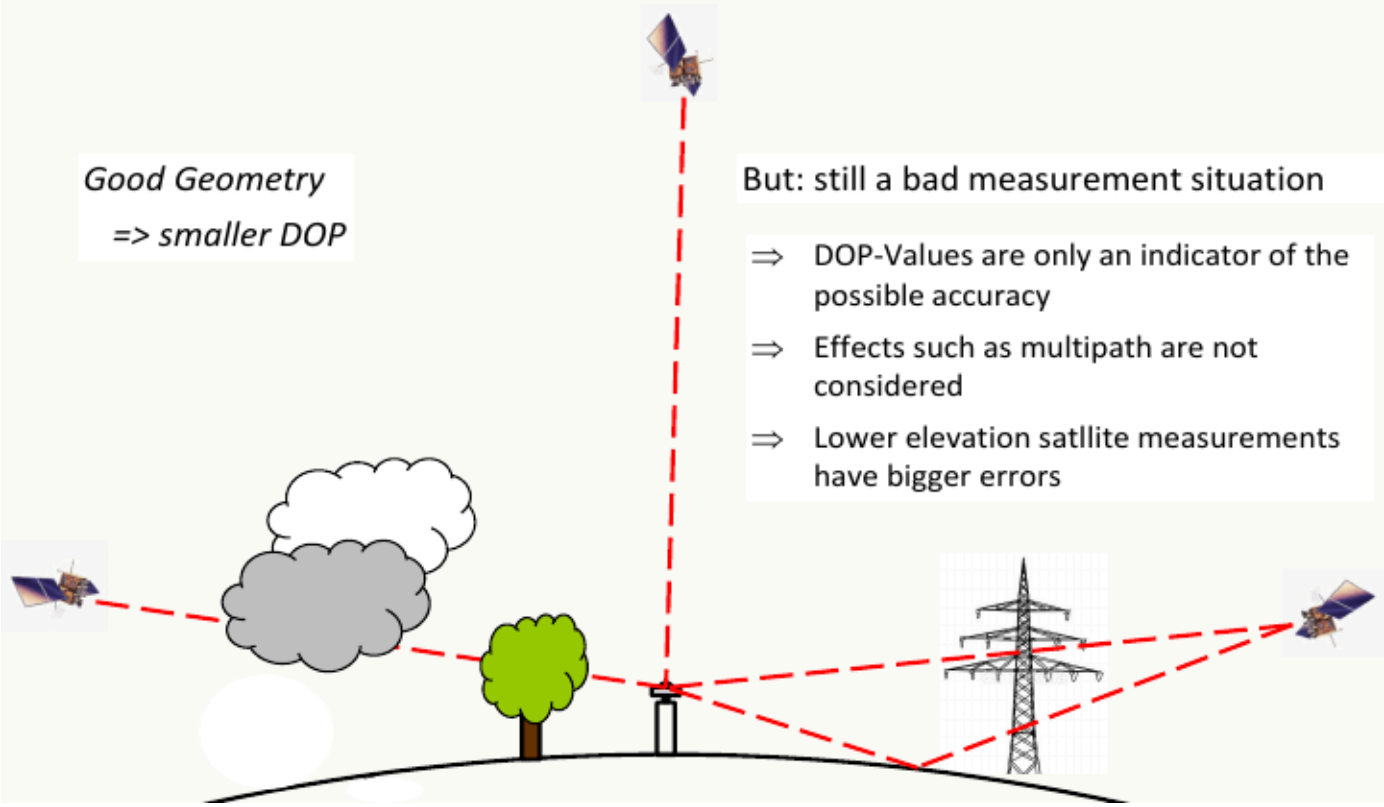
*Hofmann-W. (2008), Fig. 7.8*




- Good: PDOP<3, HDOP<2

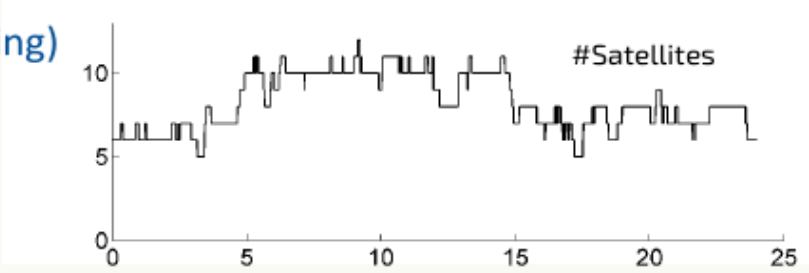

Bad Geometry
=> higher DOP

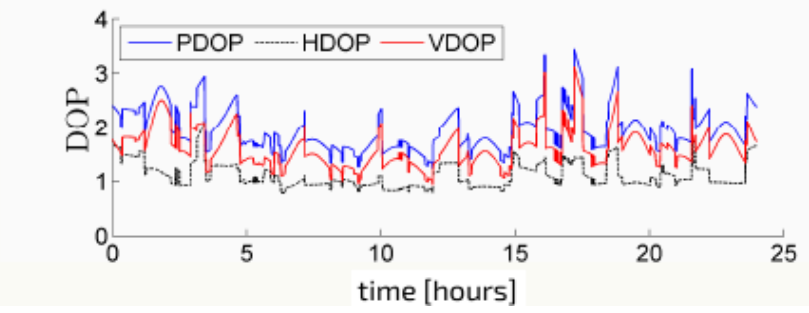

Good Geometry
=> smaller DOP

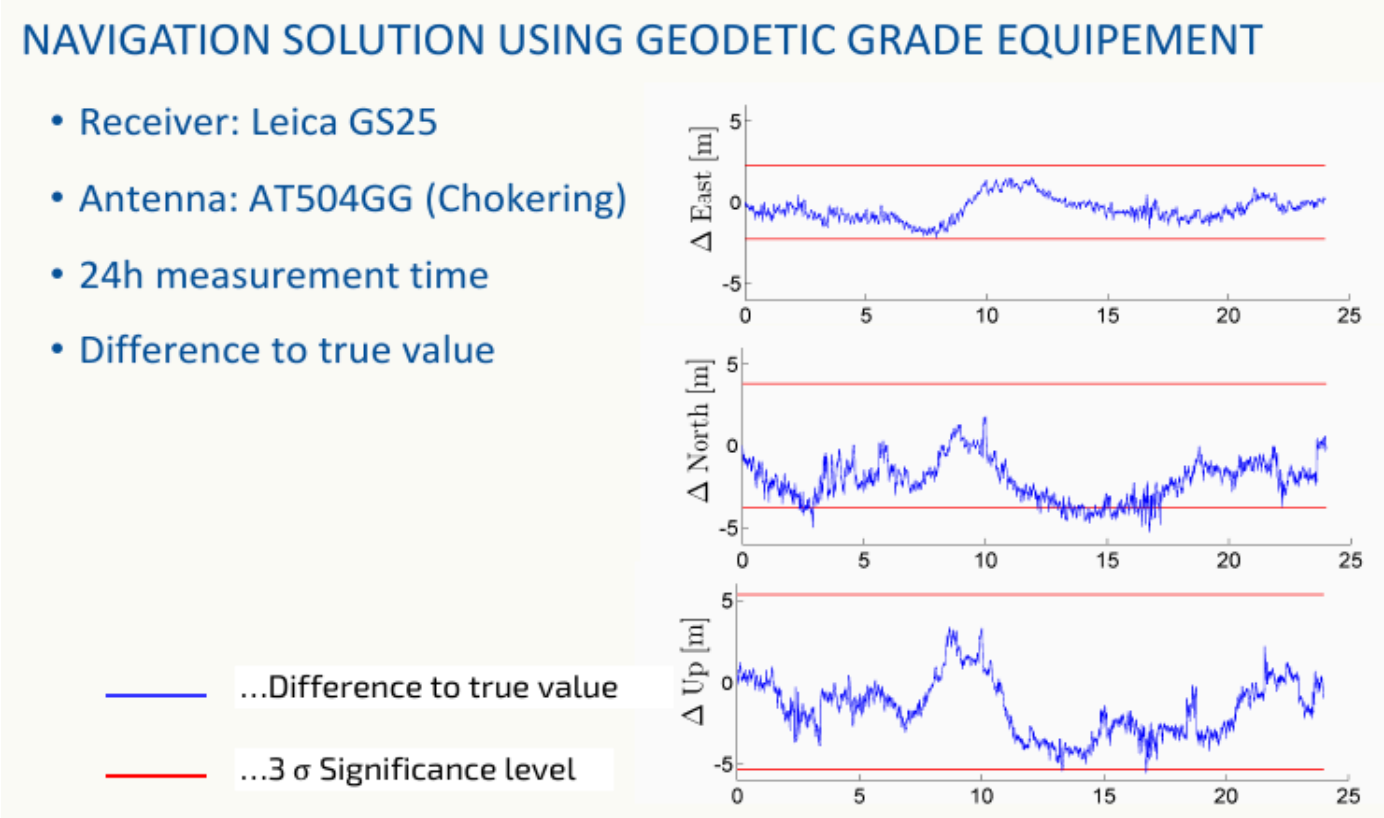



## NAVIGATION SOLUTION USING GEODETIC GRADE EQUIPEMENT

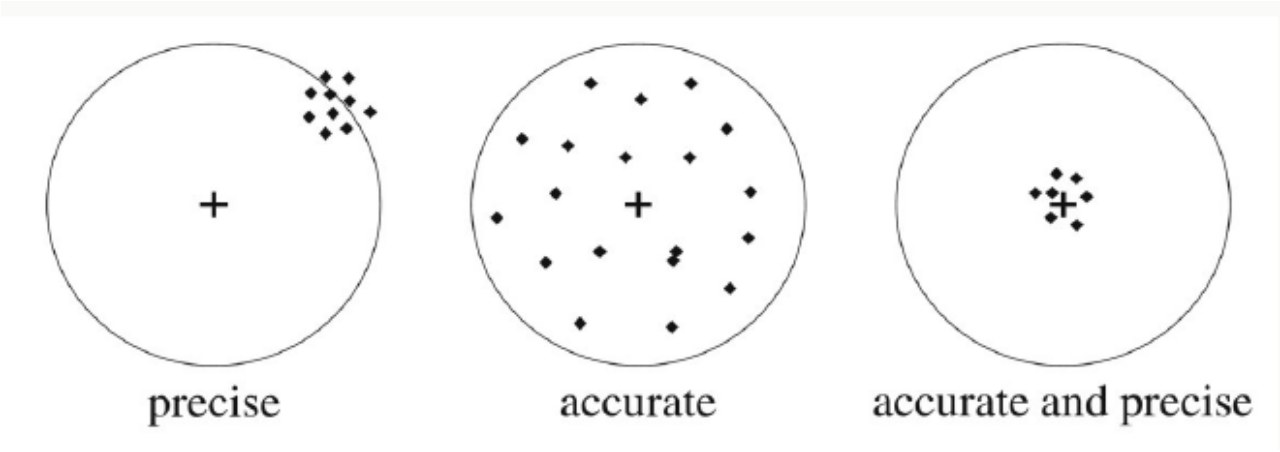
- Receiver: Leica GS25
- Antenna: AT504GG (Chokering)
- 24h measurement time
- Difference to true value

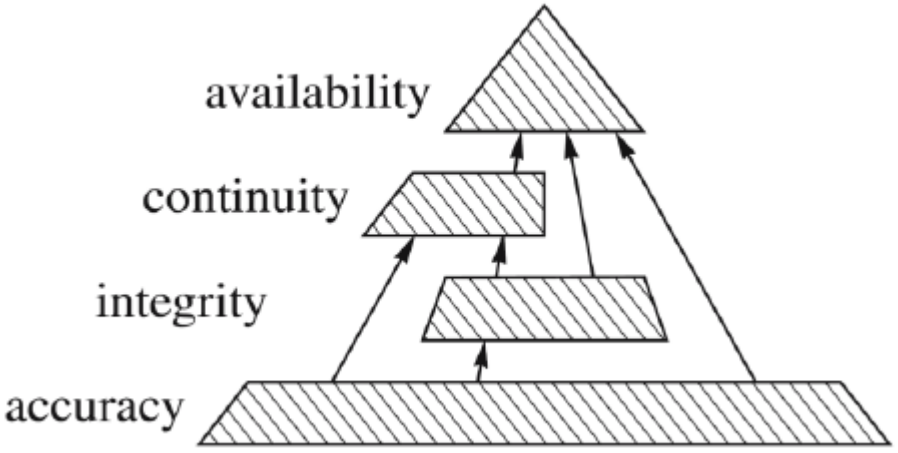





### Quality Parameters

- Accuracy
  - Usually given in meters, combined with a statistical information
  - -> e.g. The probability of the position being within a radius of x around a point is 95%
  - 

- Availability
  - Percentage of time, the system is working
    - within a certain area or
    - providing a certain accuracy, integrity or continuity
      - -> Capability of the system to provide useful information in given or special conditions
- Capacity
  - How many users can use the system at the same time
  - -> usually unlimited for GNSS

- Continuity
  - Capability of the system, to provide the needed accuracy and integrity without interruption during an operation/application
  - important for landing planes
- Coverage
  - Area, where th system operates with sufficient accuracy
  - -> GNSS usually have a global coverage, when not considering signal blocking by topography, buildings, vegetation, etc...
- Integrity
  - Capability of the system to provide warning messages/alarm if something is disfunctional
  - In GNSS:
    - Monitoring station (Ground base augmentation systems)
    - Satellite Based Augmentation Systems (SBAS)
    - Additional sensors (inertial sensors)
    - Receiver autonomous Integrity Monitoring (RAIM)
      - (e.g. statistical tests during GNSS processing, using redundant information)
- Reliability
- -> Hierarchical and correlated